

Customer Shopping Behavior Analysis

Exploratory Data Analysis Project

EncoderX Remote Internship — Data Science Batch 02

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Tools: Python, Pandas, NumPy, Matplotlib, Seaborn

1. Introduction

This project focuses on performing Exploratory Data Analysis on customer shopping behavior data. The objective is to understand customer demographics, product preferences, purchasing patterns, spending behavior, and relationships between different variables.

The analysis follows a complete EDA workflow, including data preprocessing, statistical analysis, visualization, identification of patterns, and development of actionable recommendations.

2. Dataset Overview

The dataset contains customer shopping information including demographic, product, transaction, and behavioral attributes.

Important variables include:

- Age and Gender
- Item Purchased
- Product Category
- Purchase Amount
- Location
- Season
- Review Rating
- Subscription Status
- Payment Method
- Shipping Type
- Previous Purchases
- Frequency of Purchases

The dataset contains **3,900 records** and provides sufficient information for analyzing customer purchasing behavior.

Data Cleaning & EDA

3. Data Cleaning and Preparation

The dataset was examined and prepared before performing the analysis. The preprocessing process included:

- Checking the dataset structure and data types
- Checking for missing values
- Checking for duplicate records
- Examining numerical variables
- Checking data consistency
- Identifying potential outliers
- Creating age groups for demographic analysis

After preprocessing, the cleaned dataset was saved as:

cleaned_customer_shopping_behavior.csv

4. Exploratory Analysis

The analysis focused on several major areas:

Customer Demographics:

Gender and age distributions were analyzed to understand the customer base.

Product & Category Analysis:

Products and categories were compared based on purchase frequency and average spending.

Customer Spending:

Purchase amounts were analyzed to understand customer spending patterns.

Purchase Behavior:

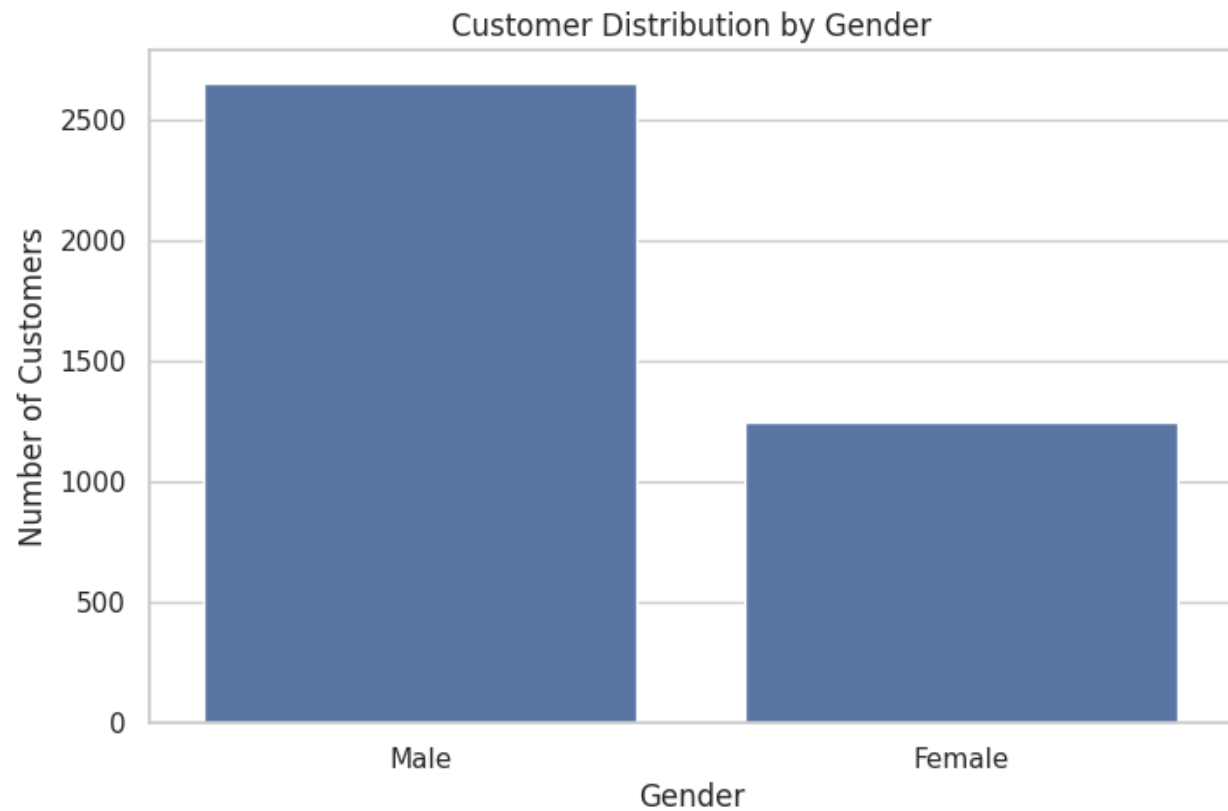
Subscription status and purchase frequency were examined.

Relationship Analysis:

The relationship between previous purchases and current purchase amount was investigated.

Visualizations & Findings\

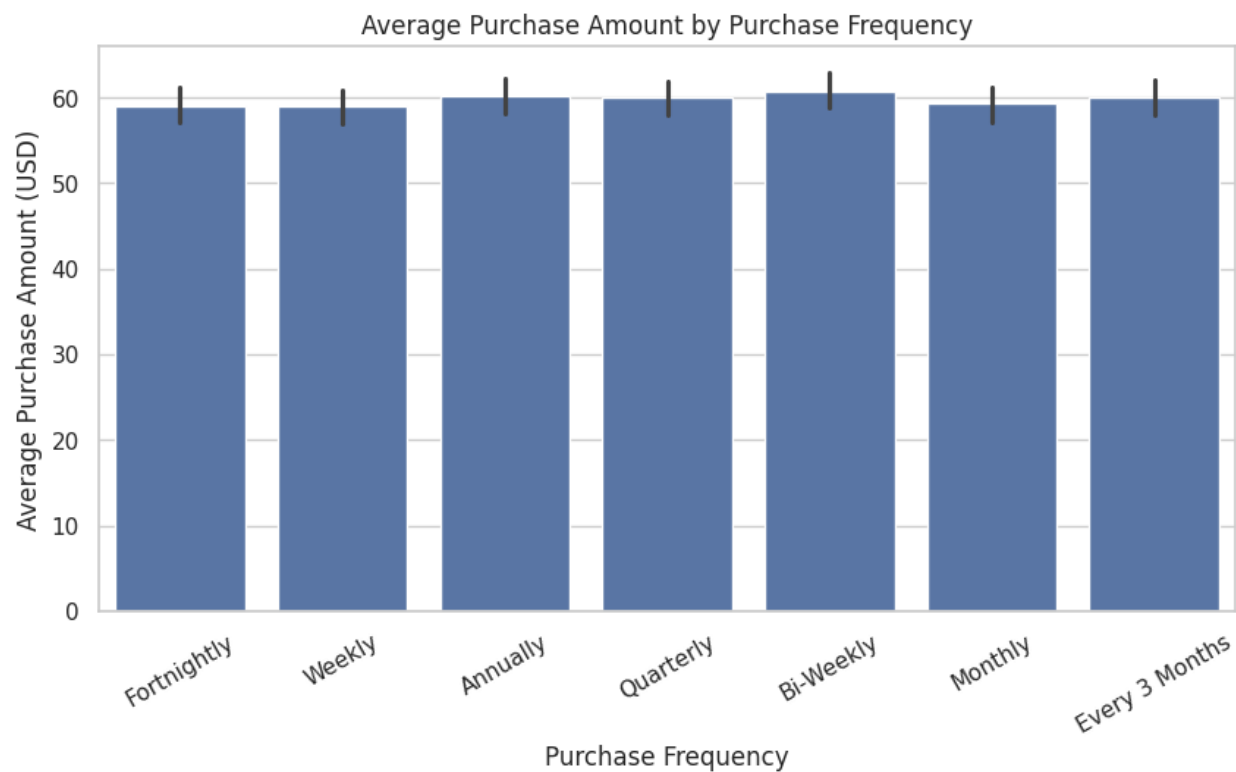
Gender Distribution



Finding:

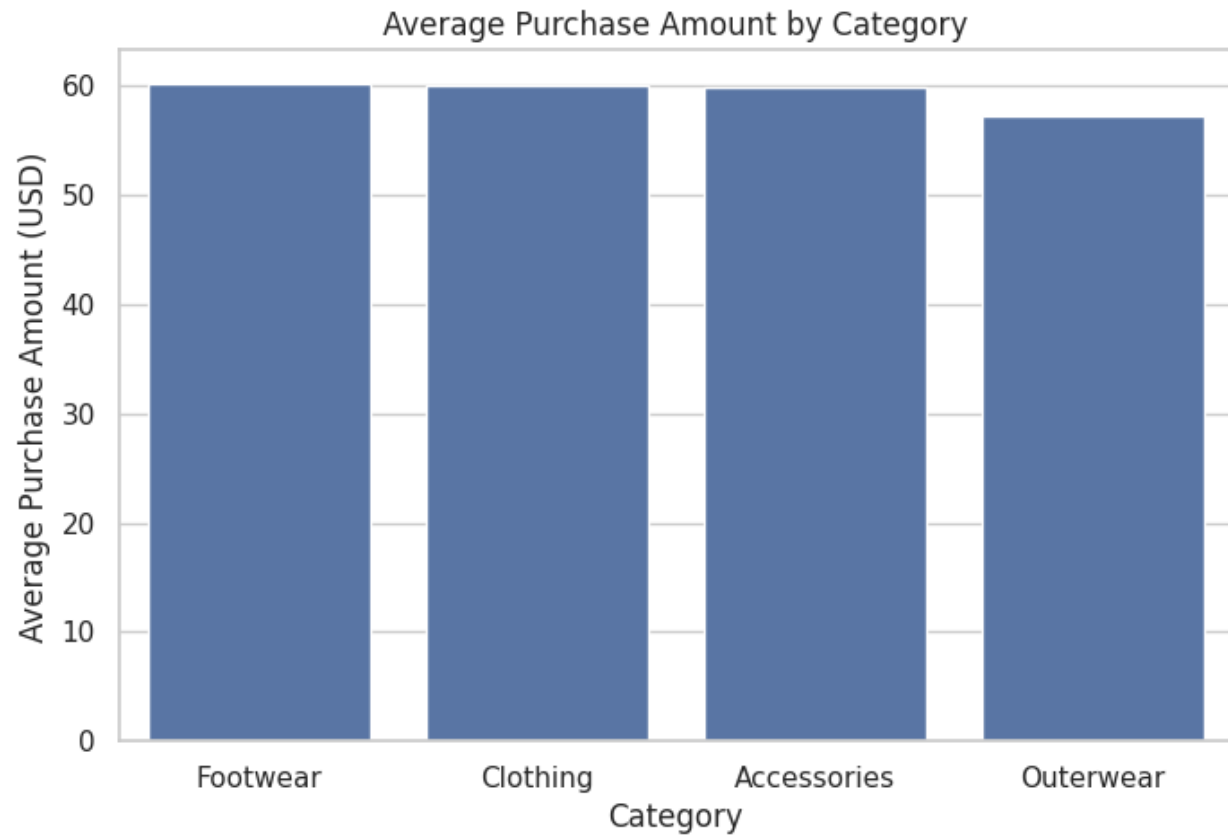
The visualization shows the distribution of purchases across genders and provides an overview of the demographic composition of the dataset.

Product & Category Analysis



Finding:
The analysis shows differences in purchase frequency and average spending between product categories.

Purchase Amount Distribution

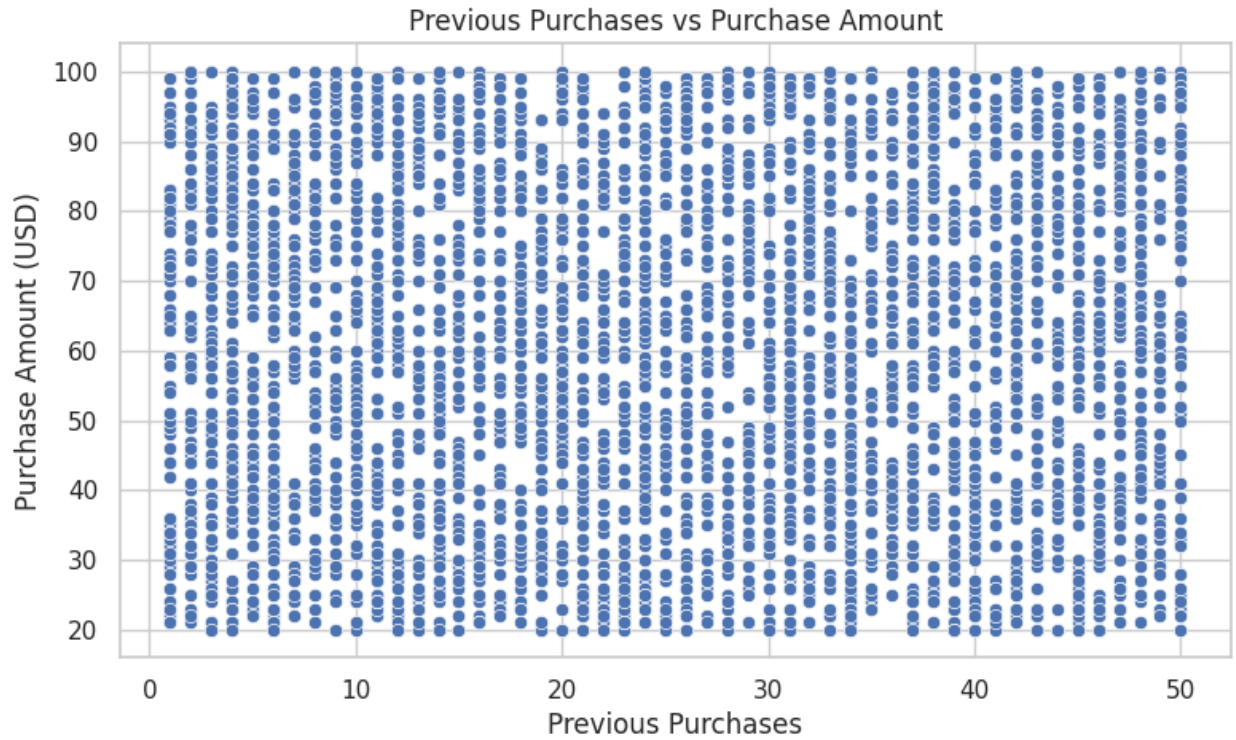


Finding:

The histogram shows the distribution of customer purchase amounts and provides an overview of the typical spending range.

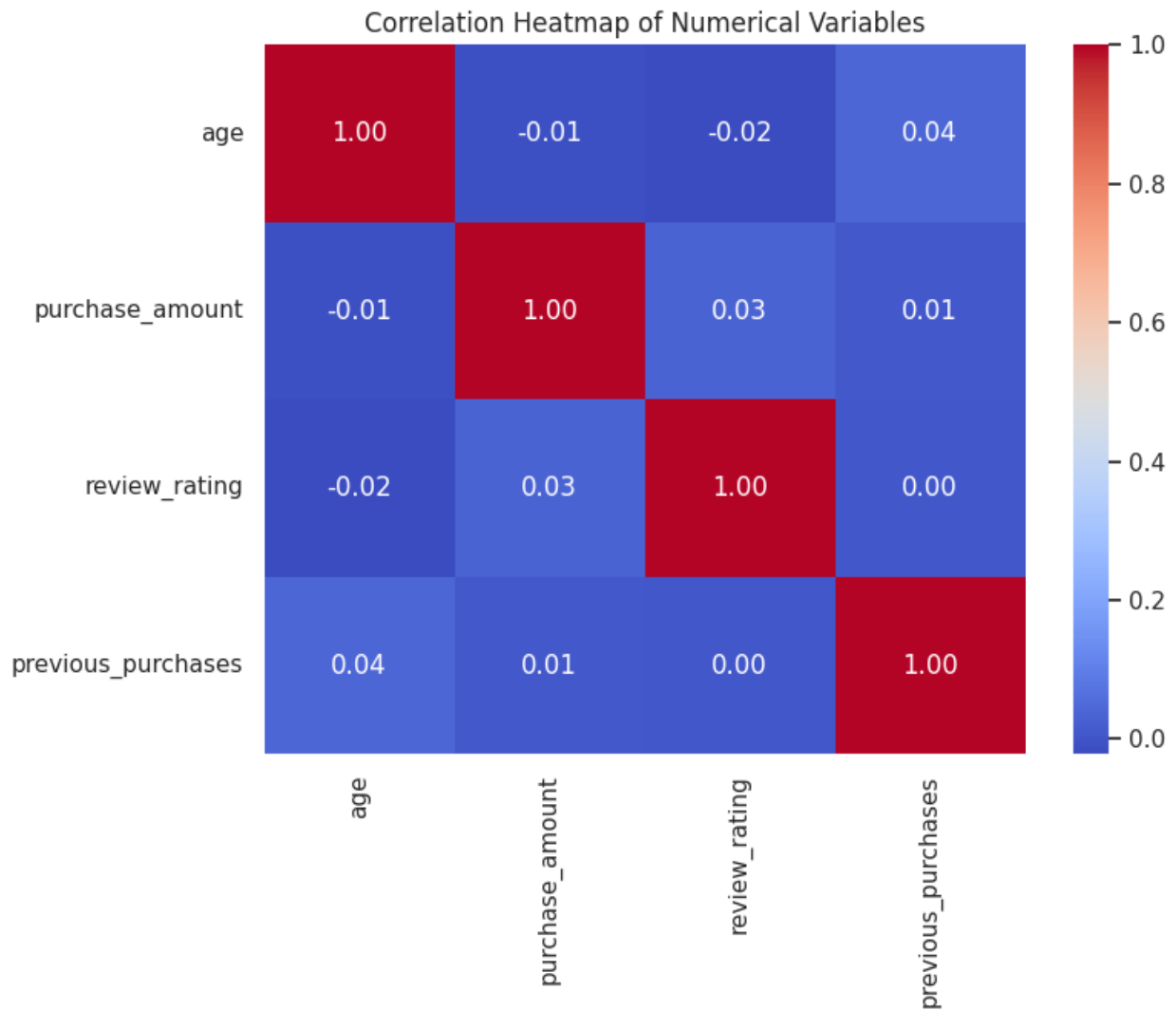
Relationship & Correlation Analysis

Previous Purchases vs Purchase Amount



The scatter plot was used to investigate whether customers with more previous purchases tend to spend more in their current purchase. The relationship appears weak, suggesting that previous purchase count alone is not a strong predictor of current spending.

Correlation Analysis



The correlation analysis included:

- Age
- Purchase Amount
- Review Rating
- Previous Purchases

The strongest observed correlation was approximately **0.04**, indicating that the numerical variables have very weak linear relationships.

This suggests that customer purchasing behavior cannot be explained effectively using a single numerical variable.

Insights & Recommendations

Key Insights

1. Product popularity varies across different categories.
2. Average spending differs between product categories.
3. Purchase frequency provides useful information about customer behavior.
4. Subscription status can be used as an additional customer segmentation factor.
5. Previous purchases have a weak relationship with current purchase amount.
6. Numerical variables show very weak linear correlations.
7. Customer behavior should be analyzed using multiple factors.

Recommendations

- Prioritize popular products and categories in marketing and inventory planning.
- Use high-spending categories for premium promotions and upselling.
- Develop customer segments based on demographics and purchasing behavior.
- Use loyalty programs to encourage repeat purchases.
- Create targeted promotions based on purchase frequency and customer characteristics.
- Combine multiple variables when making business decisions instead of relying on a single factor.

Conclusion

This analysis helped us understand customer shopping patterns, including product preferences, spending, and purchase frequency. The results show that no single numerical factor strongly determines customer spending.

Overall, using multiple customer and purchase characteristics can help businesses make better marketing and sales decisions.